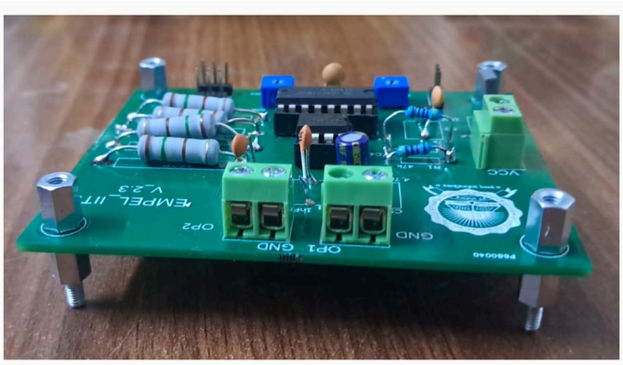
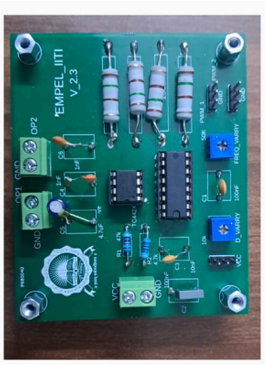
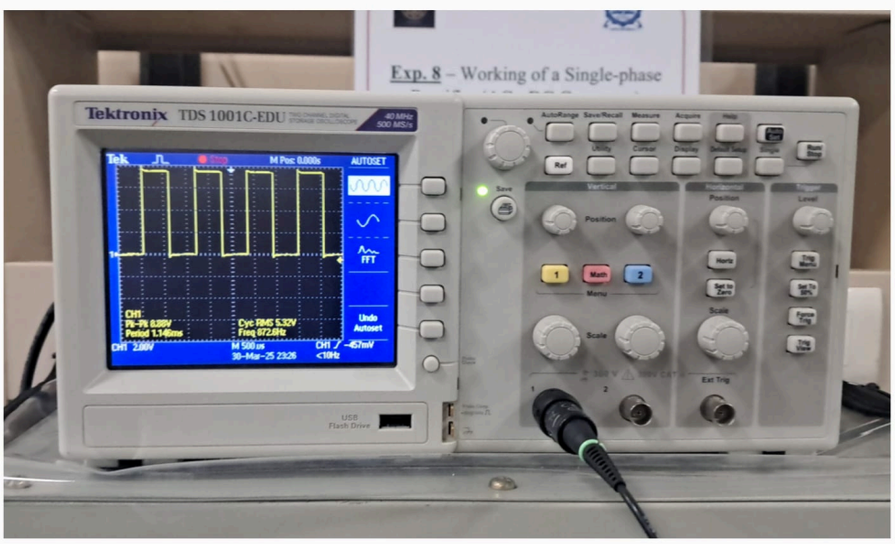
BUCK BOOST CONVERTER

Square-wave Signal constructor (Gate driver) :

- **Aim:**
  To solder the given PCB and components to create a pulse wave generator (gate-driver circuit for MOSFET).

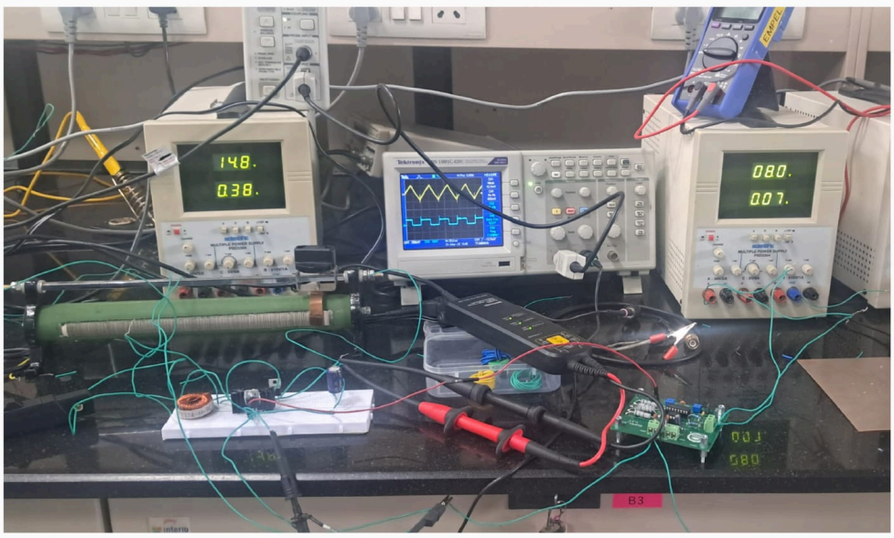




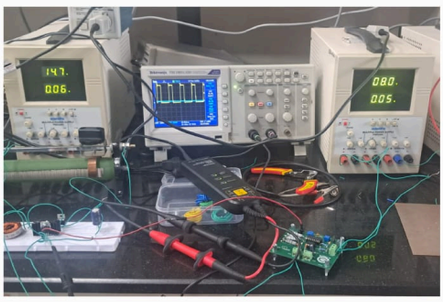
- **Key points:**
  - It was used to provide the gate signal to the MOSFET of the main circuit.
  - The potentiometers, namely D_varry and FREQ_varry, is used to change duty cycle and frequency of the pulse wave respectively.
  - The PWM_2 gives the same output as OP2.
- **Precautions and Possible reasons for failure:**
  - Check both ICs carefully before using.
  - If any of the IC's temperature is increasing, check all the connections and soldering.
  - Check the potentiometer before use.

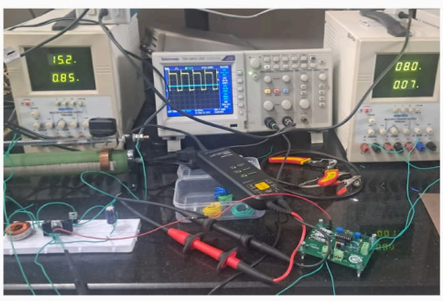
Construction of the Buck-boost circuit on the Breadboard

- **Aim:**

To construct a Buck Boost circuit on a breadboard with the given components in order to test the given circuits.







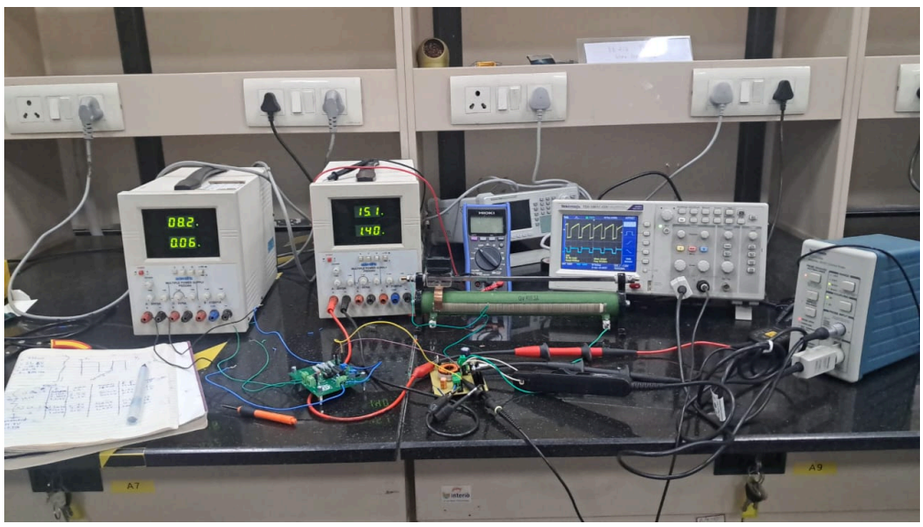
- **Key points:**
  - We made a basic buck-boost circuit on the breadboard and used the gate driver circuit to generate a pulse signal for it. The common ground was MOSFET's drain terminal.
  - The voltage across inductor was noted using a differential probe to check the waveform, which came out to be correct.
  - The switch current was observed using a current probe to check the waveform, which also came out to be correct.
  - Then, we reduced the frequency to check the DCM mode and it worked...
- **Precautions and possible reasons for failure:**
  - Check the circuit carefully before giving any input.
  - Give the gate signal first and then turn on the supply voltage.
  - Set the attenuation of differential probe and current probe accordingly.
  - Don't connect the MOSFET directly without pasting it to heatsink else it may get burnt out or/and damage the circuit.
  - In case of absurd results check if the MOSFET or diode are not sorted.

PCB construction of Buck boost circuit

- **Aim:**
  To design, construct and testing the PCB of a Buck-boost generator using the gate driver.
- **Designing:**
  - **Check out the design here : Click here**
  - **Explanation of the design:**

- left enough space on the PCB for heatsink of the MOSFET.
- For inductor voltage, we use a differential probe across the berk pins H3 (potential = supply potential) and H4 (potential = Drain's voltage).
- For switch current, we connected a wire from pin3 to pin2 of H4 as the current through it would be equal to switch current.

Testing



- **Part - A) For different values duty cycle, we find the different values of output voltage for a fixed value of input voltage, frequency and load resistance. We also observed inductor voltage and switch current. set up figure**
  - **Duty cycle = 39.5**
    frequency = 7.407 kHz
    Vpp = 10.2 V
    Input Voltage = 15.1 V

    Output Voltage Observed = -8.72 V

    Expected Voltage = -9.86 V

    % error = 11.56%
    DSO Outputs
  - **Duty cycle = 49.7**
    frequency = 7.082 kHz
    Vpp = 10.4 V
    Input Voltage = 15.1 V

    Output Voltage Observed = -13 V

    Expected Voltage = -14.92 V

    % error = 12.86 %
    DSO Outputs
  - **Duty cycle = 55.4**

frequency = 7.212 kHz
Vpp = 10.2 V
Input Voltage = 15.1 V

Output voltage observed = -16.2 V

Expected Voltage = -18.76 V

% error = 15.78%
DSO Outputs

- **Duty cycle = 61.7**

frequency = 7.226 kHz
Vpp = 10.2 V
Input Voltage = 15.1 V

Output voltage observed = -20 V

Expected Voltage = -24.32 V

% error = 17.76%
DSO Outputs

- **PART B) We have to show DCM by changing the frequency of the gate pulse.**
  Demonstration of DCM : DCM mode outputs

Matlab simulation:

Click here

Conclusion:

- Successfully designed, simulated, and fabricated a buck-boost converter.
- Gained practical insights into the operation and real-world behavior of the converter.
- Used **MATLAB Simulink** to model the converter accurately.
- Compared theoretical results with experimental data.
- Designed the PCB using EasyEDA.
- Fabricated the PCB at the Makerspace Lab, IIT Indore.
- Gained valuable hands-on experience in circuit realization.
- Deepened understanding of power electronics.
- Understood the importance of bridging simulation with practical implementation.